

# Object Detection in image processing

Using YOLO v5 in python

# Framework and Objective

## Framework:

Develop an **Object Detection System** using **YOLO v5** by training on a **custom household item dataset**. The project is implemented in **Google Colab** for model development and training, and tested **locally on PC** using Python scripts.

## Objective:

- Train a YOLOv5 model on real-world household objects captured under different conditions.
- Enable object detection capability for everyday use cases.
- Build a working model pipeline from **image acquisition** → **labeling** → **training** → **inference**.

Input Image



Dataset Gathering (Custom Images)



Labeling (Label Studio Tool)



Training (Google Colab - YOLO v5)

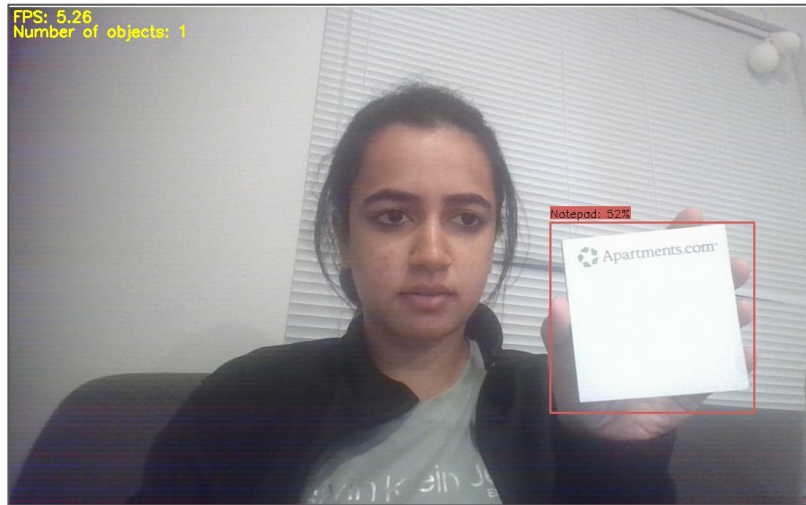


Deployment and Testing (Local PC with Python Script)

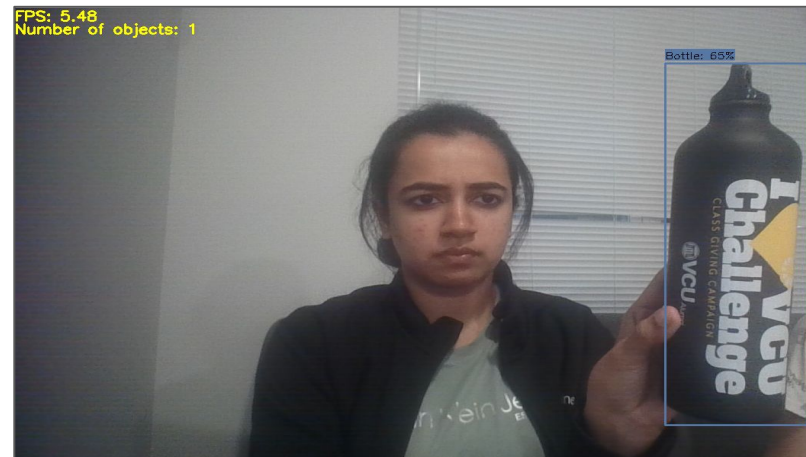


Output: Object Detected with Bounding Box and Label

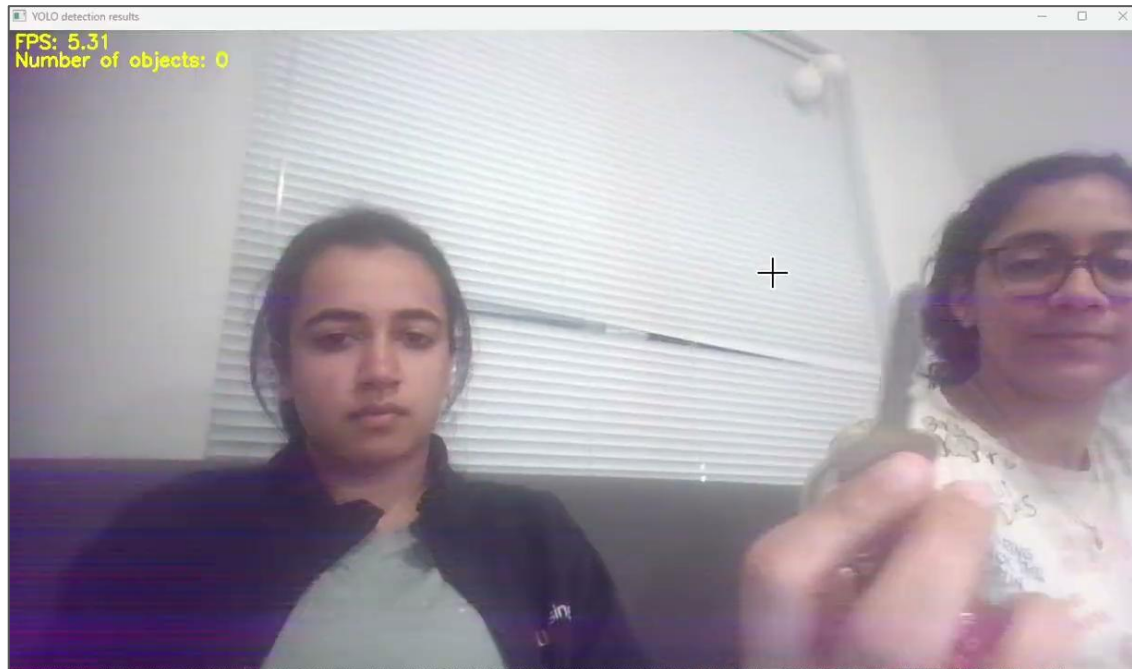
FPS: 5.26  
Number of objects: 1



FPS: 5.48  
Number of objects: 1



# Video of few Object Detections



# Featured Methods and Ideas

## Custom Dataset:

- 10–20 real images of household objects manually captured.
- Diverse conditions: lighting, background, perspective.

## Annotation:

- Used **Label Studio** (open-source labeling tool).
- Exported labeled data in YOLO format for training.

## Training Approach:

- Uploaded dataset zip to Colab.
- Trained using **YOLO v5s** lightweight model for faster results.

## Testing and Deployment:

- Model tested both in Colab and Anaconda local environment.
- Used Python scripts for local inference.

# Future Scope (Next Steps)

## Advanced Image Feature Extraction:

- **Grayscale Conversion** for reducing complexity. ( like infrared)
- **Dominant Color Detection** inside bounding boxes.
- **Size Estimation** (height, width in pixels).
- **Angle/Orientation Estimation** of objects.
- **Texture Analysis** (rough/smooth detection).

## Counting Objects:

- Detect multiple instances and count objects within frames.

## Improving Dataset:

- Add more classes, variations, and improve lighting conditions.
- Using existing pre-trained datasets so that the process is easy.

## Face Recognition Expansion:

- After object detection success, apply face detection and recognition using Deep Learning

# Current Challenges

- We trained the model using images captured manually with our camera, and even though it has given proper outputs, it is time consuming and limited to only certain number of objects.
- Runtime is very long making us wait for longer time to proceed with next steps (GPU usage is limited forcing us to use CPU session).

# Week 3 and 4

Aim: Tried using another camera, with more clarity. To overcome 2 particular challenge that we had last week:

- Trying to overcome challenge 1:  
In previous week we trained the model using images captured manually with our camera, and even though it has given proper outputs, it is time consuming and limited to only certain number of objects. ( we wanted to check if it works with a proper camera rather than phone)

We trained only wallet images because we wanted to check if its working right with one object , so that there is a possibility to do it to other objects later.

- Trying to overcome challenge 2:  
In previous week , we trained only 10 - 15 images for each object that did not give us that accurate outputs. This week we selected 74 images of wallet and trained it with the same YOLO Model. To check its accuracy.